Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2013 – Paper 2 – Solutions**

## June 2013 – Question 1 – Solution

(a) Let $\boldsymbol{p}$ and $\boldsymbol{q}$ be two propositions. Construct a truth table for the statements

(i) $\boldsymbol{p} \rightarrow \boldsymbol{q}$ [1]

The truth table for $\boldsymbol{p} \rightarrow \boldsymbol{q}$ is as follows:

| $\boldsymbol{p}$ | $\boldsymbol{q}$ | $\boldsymbol{p} \rightarrow \boldsymbol{q}$ |
|---|---|---|
| $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ |
| $F$ | $T$ | $T$ |
| $F$ | $F$ | $T$ |

(ii) $\sim(\boldsymbol{p} \wedge \boldsymbol{q})$ [2]

The truth table for $\sim(\boldsymbol{p} \wedge \boldsymbol{q})$ is as follows:

| $\boldsymbol{p}$ | $\boldsymbol{q}$ | $\boldsymbol{p} \wedge \boldsymbol{q}$ | $\sim(\boldsymbol{p} \wedge \boldsymbol{q})$ |
|---|---|---|---|
| $T$ | $T$ | $T$ | $F$ |
| $T$ | $F$ | $F$ | $T$ |
| $F$ | $T$ | $F$ | $T$ |
| $F$ | $F$ | $F$ | $T$ |

(b) A binary operator $\oplus$ is defined on a set of positive real numbers by

$$y \oplus x = y^2 + x^2 + 2y + x - 5xy$$

Solve the equation $2 \oplus x = 0$. [5]

$y \oplus x = y^2 + x^2 + 2y + x - 5xy$

$2 \oplus x = (2)^2 + x^2 + 2(2) + x - 5x(2)$

$= 4 + x^2 + 4 + x - 10x$

$= x^2 - 9x + 8$

Let $2 \oplus x = 0$.

$x^2 - 9x + 8 = 0$

$(x - 1)(x - 8) = 0$

Either $x - 1 = 0$ or $x - 8 = 0$

$x = 1$ $x = 8$

$\therefore x = 1$ and $x = 8$

(c) Use mathematical induction to prove that $5^n + 3$ is divisible by 2 for all values of $n \in \mathbb{N}$. [8]

Let $P(n)$ be the statement:

$5^n + 3$ is divisible by 2 or $2|5^n + 3, \forall n \in \mathbb{N}$

Looking at $P(1)$.

$5^1 + 3 = 8$

$= 2(4)$ which is divisible by 2.

$\therefore P(n)$ is true for $n = 1$.

Assume $P(k)$ is true.

That is, $5^k + 3$ is divisible by 2.

$\therefore 5^k + 3 = 2m$, where $m \in \mathbb{N}$

$\Rightarrow 5^k = 2m - 3$

Looking at $P(k + 1)$.

That is, $5^{k+1} + 3$ is divisible by 2.

We need to prove $P(k + 1)$.

$5^{k+1} + 3 = 5^k \cdot 5 + 3$

$= (2m - 3)(5) + 3$

$= 10m - 15 + 3$

$= 10m - 12$

$= 2(5m - 6)$ which is divisible by 2

$\therefore$ If $P(k)$ is true, $P(k+1)$ is also true.

Since $P(1)$ is true, and since $P(k+1)$ is true when $P(k)$ is true, then by the principle of mathematical induction, $P(n)$ is true for all $n$, ie.

$5^n + 3$ is divisible by 2 for all $n \in \mathbb{N}$.

Q.E.D.

(d) Let $f(x) = x^3 - 9x^2 + px + 16$.

(i) Given that $(x+1)$ is a factor of $f(x)$, show that $p = 6$. [2]

$f(x) = x^3 - 9x^2 + px + 16$

Since $(x+1)$ is a factor of $f(x)$, then by the Factor Theorem, $f(-1) = 0$.

$f(-1) = (-1)^3 - 9(-1)^2 + p(-1) + 16$

$0 = -1 - 9 - p + 16$

$0 = 6 - p$

$p = 6$

$\therefore p = 6$

Q.E.D.

(ii) Factorise $f(x)$ completely. [4]

$f(x) = x^3 - 9x^2 + px + 16$

$f(x) = x^3 - 9x^2 + 6x + 16$ $[\because p = 6]$

We are given that $(x + 1)$ is a factor of $f(x)$.

By long division,

$$
\begin{array}{r|l}
 & x^2 - 10x + 16 \\
\hline
x + 1 & x^3 - 9x^2 + 6x + 16 \\
 & x^3 + x^2 \\
\hline
 & -10x^2 + 6x \\
 & -10x^2 - 10x \\
\hline
 & 16x + 16 \\
 & 16x + 16 \\
\hline
 & 0 \\
\hline
\end{array}
$$

Now,

$f(x) = (x + 1)(x^2 - 10x + 16)$

$= (x + 1)(x - 8)(x - 2)$

$\therefore$ The factors of $f(x)$ are $(x + 1)$, $(x - 8)$ and $(x - 2)$.

(iii) Hence, or otherwise, solve $f(x) = 0$. [3]

$f(x) = (x+1)(x-8)(x-2)$

Let $f(x) = 0$. Then,

$(x+1)(x-8)(x-2) = 0$

Either $x + 1 = 0$ or $x - 8 = 0$ or $x - 2 = 0$

$x = -1$ $x = 8$ $x = 2$

$\therefore x = -1, x = 2$ and $x = 8$

**Total: 25 marks**

## June 2013 – Question 2 – Solution

(a) Let $A = \{x : x \in \mathbb{R}, x \geq 1\}$

A function $f : A \to \mathbb{R}$ is defined as $f(x) = x^2 - x$. Show that $f$ is one to one. [7]

$f(x) = x^2 - x$

Required to show that $f$ is one to one.

$f(a) = a^2 - a$

$f(b) = b^2 - b$

Let $f(a) = f(b)$. Then,

$$a^2 - a = b^2 - b$$

$$a^2 - b^2 - a + b = 0$$

$$(a - b)(a + b) - (a - b) = 0$$

$$(a - b)(a + b - 1) = 0$$

Either $a - b = 0$ or $a + b - 1 = 0$

$a = b$ $a + b = 1$ But $a, b \geq 1$.

So, this is not possible.

$\therefore$ Since $f(a) = f(b)$ implies $a = b$, then $f$ is one-to-one.

Q.E.D.

<u>Alternative Method:</u>

$f(x) = x^2 - x$

$f(x) = x(x-1)$

When $f(x) = 0$,

$x(x-1) = 0$

Either $x = 0$ or $x - 1 = 0$

$x = 1$

$\therefore$ The roots of the quadratic are $(0, 0)$ and $(1, 0)$.

The $x$-coordinate of $f(x)$ is

$x = -\frac{b}{2a}$

$= -\frac{(-1)}{2(1)}$

$= \frac{1}{2}$

When $x = \frac{1}{2}$,

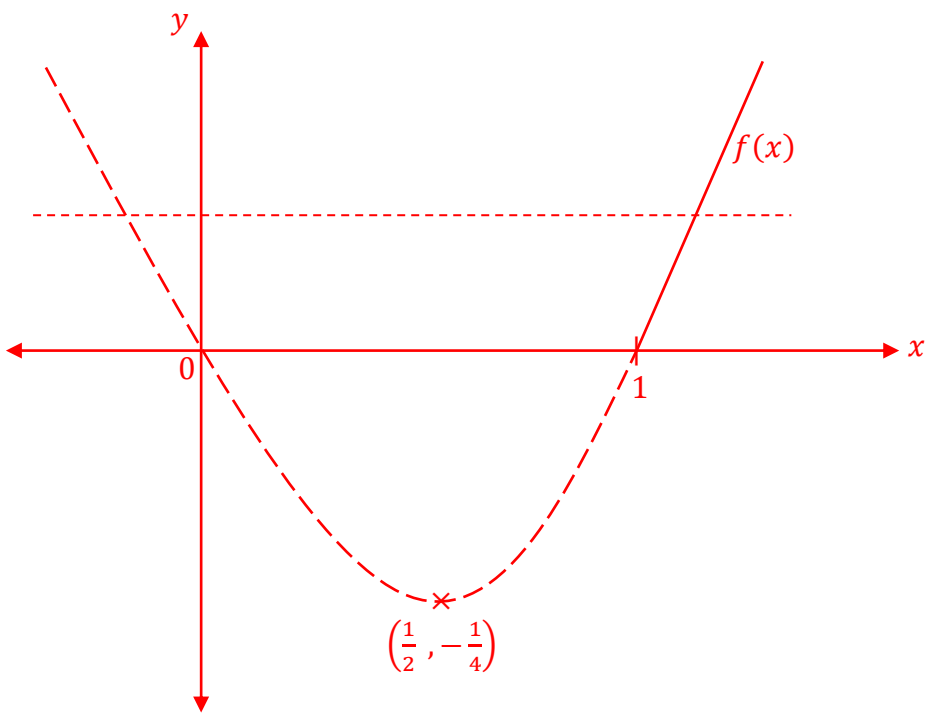
$y = \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)$

$= \frac{1}{4} - \frac{1}{2}$

$= -\frac{1}{4}$

$\therefore$ The minimum point is $\left(\frac{1}{2}, -\frac{1}{4}\right)$.

The curve is as follows:




If the function $f$ is one-to-one, then $\forall y \in \mathbb{R}$ will map onto at most one $x \in \mathbb{R}$.

Since the domain is restricted, any line parallel to the $x$-axis will cut the graph at most once.

Hence, $f$ is one-to-one.

Q.E.D.

(b) Let $f(x) = 3x + 2$ and $g(x) = e^{2x}$.

(i) Find

(a) $f^{-1}(x)$ and $g^{-1}(x)$ [4]

$f(x) = 3x + 2$

Let $y = f(x)$.

$y = 3x + 2$

Interchanging variables $x$ and $y$:

$x = 3y + 2$

Making $y$ the subject of the formula:

$x - 2 = 3y$

$\frac{x-2}{3} = y$

$\therefore f^{-1}(x) = \frac{x-2}{3}$

$g(x) = e^{2x}$

Let $y = g(x)$.

$y = e^{2x}$

Interchanging variables $x$ and $y$:

$x = e^{2y}$

Making $y$ the subject of the formula:

$\ln x = \ln e^{2y}$

$\ln x = 2y \ln e$

$\ln x = 2y$

$\frac{1}{2}\ln x = y$

$\therefore g^{-1}(x) = \frac{1}{2}\ln x$

(b) $f[g(x)]$ (or $f \circ g(x)$) [1]

$f[g(x)] = f(e^{2x})$

$= 3e^{2x} + 2$

(ii) Show that $(f \circ g)^{-1}(x) = g^{-1}(x) \circ f^{-1}(x)$. [5]

Required to show that $(f \circ g)^{-1}(x) = g^{-1}(x) \circ f^{-1}(x)$.

Taking L.H.S:

$fg(x) = 3e^{2x} + 2$

Let $y = fg(x)$.

$y = 3e^{2x} + 2$

Interchanging variables $x$ and $y$:

$x = 3e^{2y} + 2$

Making $y$ the subject of the formula:

$$x - 2 = 3e^{2y}$$

$$\frac{x-2}{3} = e^{2y}$$

$$\ln\left(\frac{x-2}{3}\right) = \ln e^{2y}$$

$$\ln\left(\frac{x-2}{3}\right) = 2y \ln e$$

$$\ln\left(\frac{x-2}{3}\right) = 2y$$

$$\frac{1}{2}\ln\left(\frac{x-2}{3}\right) = y$$

So, we have $(f \circ g)^{-1}(x) = \frac{1}{2}\ln\left(\frac{x-2}{3}\right)$.

Taking R.H.S:

$$g^{-1}(x) \circ f^{-1}(x) = g^{-1}[f^{-1}(x)]$$

$$= g^{-1}\left(\frac{x-2}{3}\right)$$

$$= \frac{1}{2}\ln\left(\frac{x-2}{3}\right)$$

Since $(f \circ g)^{-1}(x) = \frac{1}{2}\ln\left(\frac{x-2}{3}\right)$ and $g^{-1}(x) \circ f^{-1}(x) = \frac{1}{2}\ln\left(\frac{x-2}{3}\right)$, then

$(f \circ g)^{-1}(x) = g^{-1}(x) \circ f^{-1}(x)$

Q.E.D.

(c) Solve the following:

(i) $3x^2 + 4x + 1 \leq 5$ [4]

$$3x^2 + 4x + 1 \leq 5$$

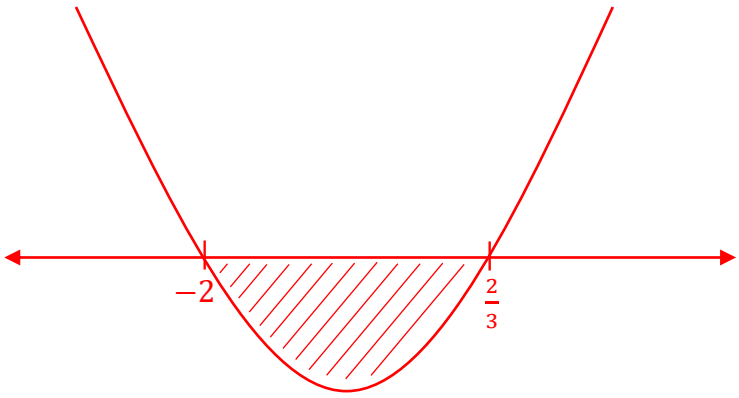
$$3x^2 + 4x + 1 - 5 \leq 0$$

$$3x^2 + 4x - 4 \leq 0$$

$$(3x - 2)(x + 2) \leq 0$$

The critical values are $x = \frac{2}{3}$ and $x = -2$.

The sketch of the graph of $y = 3x^2 + 4x - 4$ is shown below.




Since we are solving $(3x - 2)(x + 2) \leq 0$, we look where the graph is negative.

The graph is negative for $-2 \leq x \leq \frac{2}{3}$.

$\therefore$ The solution set is $\left\{x: -2 \leq x \leq \frac{2}{3}\right\}$.

(ii) $|x+2| = 3x+5$ [4]

$|x+2| = 3x+5$

By definition:

| | | |
|---|---|---|
| $x+2 = 3x+5$ | or | $x+2 = -(3x+5)$ |
| $2-5 = 3x-x$ | | $x+2 = -3x-5$ |
| $-3 = 2x$ | | $3x+x = -2-5$ |
| $x = -\frac{3}{2}$ | | $4x = -7$ |
| | | $x = -\frac{7}{4}$ |

<u>Checking solutions:</u>

Consider $x = -\frac{3}{2}$.

L.H.S. $= \left|\left(-\frac{3}{2}\right)+2\right|$

$= \left|\frac{1}{2}\right|$

$= \frac{1}{2}$

R.H.S. $= 3\left(-\frac{3}{2}\right)+5$

$= -\frac{9}{2}+5$

$= \frac{1}{2}$

So, $x = -\frac{3}{2}$ is a solution.

Consider $x = -\frac{7}{4}$.

L.H.S. $= \left|\left(-\frac{7}{4}\right) + 2\right|$

$= \left|\frac{1}{4}\right|$

$= \frac{1}{4}$

R.H.S. $= 3\left(-\frac{7}{4}\right) + 5$

$= -\frac{21}{4} + 5$

$= -\frac{1}{4}$

Since L.H.S. $\neq$ R.H.S., then $x = -\frac{7}{4}$ is not a solution.

$\therefore x = -\frac{3}{2}$

**Total: 25 marks**

## June 2013 – Question 3 – Solution

(a) (i) Show that $\sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta}$. [4]

Required to show that $\sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta}$.

Taking R.H.S:

$$\frac{2\tan\theta}{1+\tan^2\theta} = \frac{2\tan\theta}{\sec^2\theta}$$

$$= 2\tan\theta \div \sec^2\theta$$

$$= \frac{2\sin\theta}{\cos\theta} \div \frac{1}{\cos^2\theta}$$

$$= \frac{2\sin\theta}{\cos\theta} \times \frac{\cos^2\theta}{1}$$

$$= 2\sin\theta\cos\theta$$

$$= \sin 2\theta$$

$$= \text{R.H.S.}$$

$$\therefore \sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta}$$

Q.E.D.

(ii) Hence, or otherwise, solve $\sin 2\theta - \tan\theta = 0$ for $0 \le \theta \le 2\pi$. [8]

$\sin 2\theta - \tan\theta = 0$

Now,

$\sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta}$

So, we have,

$\frac{2\tan\theta}{1+\tan^2\theta} - \tan\theta = 0$

$[\times (1+\tan^2\theta)]$

$2\tan\theta - \tan\theta\,(1+\tan^2\theta) = 0$

$2\tan\theta - \tan\theta - \tan^3\theta = 0$

$\tan\theta - \tan^3\theta = 0$

$\tan^3\theta - \tan\theta = 0$

$\tan\theta\,(\tan^2\theta - 1) = 0$

$\tan\theta\,(\tan\theta + 1)(\tan\theta - 1) = 0$

When $\tan\theta = 0$

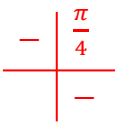
$\theta = 0, \pi, 2\pi$

When $\tan\theta + 1 = 0$

$\tan\theta = -1$

$\theta = \pi - \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$

$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$

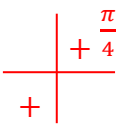


When $\tan\theta - 1 = 0$

$$\tan\theta = 1$$

$$\theta = \frac{\pi}{4}, \pi + \frac{\pi}{4}$$

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}$$




$$\therefore \theta = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{7\pi}{4}, 2\pi$$

(b) (i) Express $f(\theta) = 3\cos\theta - 4\sin\theta$ in the form $r\cos(\theta + \alpha)$ where

$r > 0$ and $0° \leq a \leq \frac{\pi}{2}$. [4]

$$f(\theta) = 3\cos\theta - 4\sin\theta$$

$$3\cos\theta - 4\sin\theta = r\cos(\theta + \alpha)$$

$$3\cos\theta - 4\sin\theta = r\cos\theta\cos\alpha - r\sin\theta\sin\alpha$$

Equating coefficients of $\sin\theta$ and $\cos\theta$ gives:

$3 = r\cos\alpha \quad \rightarrow$ Equation 1

$4 = r\sin\alpha \quad \rightarrow$ Equation 2

Equation 2 ÷ Equation 1 gives:

$$\frac{r\sin\alpha}{r\cos\alpha} = \frac{4}{3}$$

$$\tan\alpha = \frac{4}{3}$$

$$\alpha = \tan^{-1}\left(\frac{4}{3}\right)$$

$\alpha = 0.927^c$ (to 3 significant figures)

And,

$r = \sqrt{3^2 + 4^2}$

$= \sqrt{9 + 16}$

$= \sqrt{25}$

$= 5$

$\therefore 3\cos\theta - 4\sin\theta = 5\cos(\theta + 0.927^c)$

which is of the form $r\cos(\theta + \alpha)$

where $r = 5$ and $\alpha = 0.927^c$

(ii) Hence, find

(a) the maximum value of $f(\theta)$ [2]

$f(\theta) = 3\cos\theta - 4\sin\theta$

$f(\theta) = 5\cos(\theta + 0.927^c)$

Now,

$-1 \leq \cos\theta \leq 1$

$-1 \leq \cos(\theta + 0.927^c) \leq 1$

$-5 \leq 5\cos(\theta + 0.927^c) \leq 5$

$\therefore$ The maximum value of $f(\theta)$ is 5.

(b) the minimum value of $\frac{1}{8+f(\theta)}$ [2]

$f(\theta) = 5\cos(\theta + 0.927^c)$

Now,

$$-1 \leq \cos\theta \leq 1$$

$$-1 \leq \cos(\theta + 0.927^c) \leq 1$$

$$-5 \leq 5\cos(\theta + 0.927^c) \leq 5$$

$$-5 \leq f(\theta) \leq 5$$

$$8 - 5 \leq 8 + f(\theta) \leq 8 + 5$$

$$3 \leq 8 + f(\theta) \leq 13$$

$$\frac{1}{3} \geq \frac{1}{8+f(\theta)} \geq \frac{1}{13}$$

$$\frac{1}{13} \leq \frac{1}{8+f(\theta)} \leq \frac{1}{3}$$

$\therefore$ The minimum value of $\frac{1}{8+f(\theta)}$ is $\frac{1}{13}$.

(iii) Given that the sum of the angles $A$, $B$ and $C$ of a triangle is $\pi$ radians, show that

(a) $\sin A = \sin(B + C)$ [3]

We are given that the sum of the angles $A$, $B$ and $C$ of a triangle is $\pi$ radians.

Also, recall that

$\sin(P - Q) = \sin P \cos Q - \cos P \sin Q$

Now,

$A + B + C = \pi$

$A = \pi - (B + C)$

Taking sin on both sides gives:

$\sin A = \sin[\pi - (B + C)]$

$\sin A = \sin \pi \cos(B + C) - \cos \pi \sin(B + C)$

$\sin A = 0 \cdot \cos(B + C) - (-1) \cdot \sin(B + C)$

$\sin A = 0 + \sin(B + C)$

$\sin A = \sin(B + C)$

Q.E.D.

(b) $\sin A + \sin B + \sin C = \sin(A + B) + \sin(B + C) + \sin(A + C)$ [2]

Using the result from part (a),

$A = \pi - (B + C)$

$\sin A = \sin(B + C)$

Similarly,

$B = \pi - (A + C)$

$\sin B = \sin(A + C)$

And,

$C = \pi - (A + B)$

$\sin C = \sin(A + B)$

Taking L.H.S:

$$\sin A + \sin B + \sin C = \sin(B + C) + \sin(A + C) + \sin(A + B)$$

$$= \sin(A + B) + \sin(B + C) + \sin(A + C)$$

$$= \text{R.H.S.}$$

$$\therefore \sin A + \sin B + \sin C = \sin(A + B) + \sin(B + C) + \sin(A + C)$$

Q.E.D.

**Total: 25 marks**

## June 2013 – Question 4 – Solution

(a) A circle $C$ is defined by the equation $x^2 + y^2 - 6x - 4y + 4 = 0$.

(i) Show that the centre and the radius of the circle, $C$, are $(3, 2)$ and 3, respectively. [3]

$$x^2 + y^2 - 6x - 4y + 4 = 0$$

$$x^2 - 6x + y^2 - 4y = -4$$

$$x^2 - 6x + 9 + y^2 - 4y + 4 = -4 + 4 + 9$$

$$(x - 3)^2 + (y - 2)^2 = 9$$

which is of the form $(x - a)^2 + (y - b)^2 = r^2$,

where $C(3, 2)$ and radius, $r = 3$ units.

Q.E.D.

(ii) (a) Find the equation of the normal to the circle $C$ at the point $(6, 2)$. [3]

The points are $C(3, 2)$ and $(6, 2)$.

$$\text{Gradient} = \frac{y_2 - y_2}{x_2 - x_1}$$

$$= \frac{2-2}{6-3}$$

$$= \frac{0}{3}$$

$$= 0$$

Since the gradient is 0, then the equation of the normal is a horizontal line.

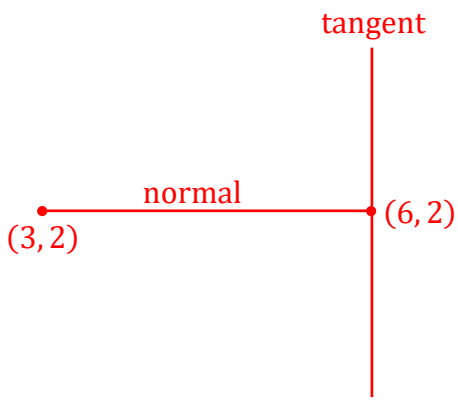
The points $(3, 2)$ and $(6, 2)$ lie on the horizontal.

$\therefore$ The equation of the normal is: $y = 2$

(b) Show that the tangent to the circle at the point $(6, 2)$ is parallel to the $y$-axis. [3]

Equation of the normal is $y = 2$.

Consider the sketch below:




$\therefore$ The equation of the tangent is $x = 6$ which is parallel to the $y$-axis.

Q.E.D.

(b) Show that the Cartesian equation of the curve that has the parametric equations

$$x = t^2 + t\,, \quad y = 2t - 4$$

is $4x = y^2 + 10y + 24$. [4]

$x = t^2 + t$ $\rightarrow$ Equation 1

$y = 2t - 4$ $\rightarrow$ Equation 2

Rearranging Equation 2 gives:

$2t = y + 4$

$t = \frac{y+4}{2}$ $\rightarrow$ Equation 3

Substituting Equation 3 into Equation 1 gives:

$$x = \left(\frac{y+4}{2}\right)^2 + \left(\frac{y+4}{2}\right)$$

$$x = \frac{(y+4)^2}{4} + \frac{y+4}{2}$$

$(\times 4)$

$$4x = (y+4)^2 + 2(y+4)$$

$$4x = y^2 + 8y + 16 + 2y + 8$$

$$4x = y^2 + 10y + 24$$

$\therefore$ The Cartesian equation of the curve is: $4x = y^2 + 10y + 24$

Q.E.D.

(c) The points $A(3, -1, 2)$, $B(1, 2, -4)$ and $C(-1, 1, -2)$ are three vertices of a parallelogram $ABCD$.

(i) Express the vectors $\overrightarrow{AB}$ and $\overrightarrow{BC}$ in the form $x\hat{\boldsymbol{i}} + y\hat{\boldsymbol{j}} + z\hat{\boldsymbol{k}}$. [3]

The points are $A(3, -1, 2)$, $B(1, 2, -4)$ and $C(-1, 1, -2)$.

So, the position vectors are:

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

$$\overrightarrow{OB} = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix}$$

$$\overrightarrow{OC} = \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix}$$

Now,

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$= \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} - \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1-3 \\ 2-(-1) \\ -4-2 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ 3 \\ -6 \end{pmatrix}$$

$= -2\hat{\boldsymbol{\imath}} + 3\hat{\boldsymbol{\jmath}} - 6\hat{\boldsymbol{k}}$ which is in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}$,

where $x = -2, y = 3$ and $z = -6$.

And,

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB}$$

$$= \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix}$$

$$= \begin{pmatrix} -1-1 \\ 1-2 \\ -2-(-4) \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$$

$= -2\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + 2\hat{\boldsymbol{k}}$ which is in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}$,

where $x = -2, y = -1$ and $z = 2$.

(ii) Show that the vector $\boldsymbol{r} = -16\hat{\boldsymbol{j}} - 8\hat{\boldsymbol{k}}$ is perpendicular to the plane through $A$, $B$ and $C$. [5]

$\boldsymbol{r} = -16\hat{\boldsymbol{j}} - 8\hat{\boldsymbol{k}}$

$$\boldsymbol{r} = \begin{pmatrix} 0 \\ -16 \\ -8 \end{pmatrix}$$

The dot product of perpendicular vectors is equal to zero.

$$\boldsymbol{r} \cdot \overrightarrow{AB} = \begin{pmatrix} 0 \\ -16 \\ -8 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 3 \\ -6 \end{pmatrix}$$

$$= (0)(-2) + (-16)(3) + (-8)(-6)$$

$$= 0 - 48 + 48$$

$$= 0$$

Also,

$$\boldsymbol{r} \cdot \overrightarrow{BC} = \begin{pmatrix} 0 \\ -16 \\ -8 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$$

$$= (0)(-2) + (-16)(-1) + (-8)(2)$$

$$= 0 + 16 - 16$$

$$= 0$$

Since $\boldsymbol{r}$ is perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{BC}$, then it is perpendicular to the plane through $A, B$ and $C$.

$\therefore \boldsymbol{r} = -16\hat{\boldsymbol{j}} - 8\hat{\boldsymbol{k}}$ is perpendicular to the plane through $A, B$ and $C$.

Q.E.D.

(iii) Hence, find the Cartesian equation of the plane through $A$, $B$ and $C$. [4]

Let $\boldsymbol{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ , $\boldsymbol{a} = \overrightarrow{OA} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ and $\boldsymbol{n} = \begin{pmatrix} 0 \\ -16 \\ -8 \end{pmatrix}$.

The Cartesian equation of the plane is

$$\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -16 \\ -8 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -16 \\ -8 \end{pmatrix}$$

$$(x)(0) + (y)(-16) + (z)(-8) = (3)(0) + (-1)(-16) + (2)(-8)$$

$$0 - 16y - 8z = 0 + 16 - 16$$

$$-16y - 8z = 0$$

$$16y + 8z = 0$$

$$2y + z = 0$$

$\therefore$ The Cartesian equation of the plane through $A$, $B$ and $C$ is $2y + z = 0$.

**Total: 25 marks**

June 2013 – Question 5 – Solution

(a) A function $f(x)$ is defined as $f(x) = \begin{cases} x+2, & x<2 \\ x^2, & x>2 \end{cases}$.

(i) Find $\lim_{x \to 2} f(x)$. [4]

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} (x+2)$$

$$= 2+2$$

$$= 4$$

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} x^2$$

$$= (2)^2$$

$$= 4$$

Since $\lim_{x \to 2^-} f(x) = 4 = \lim_{x \to 2^+} f(x)$, then $\lim_{x \to 2} f(x) = 4$.

(ii) Determine whether $f(x)$ is continuous at $x = 2$. **Give a reason for your answer.** [2]

Since $f(x)$ is not defined at $x = 2$, then $f(x)$ is not continuous at $x = 2$.

(b) Let $y = \frac{x^2+2x+3}{(x^2+2)^3}$. Show that $\frac{dy}{dx} = \frac{-4x^3-10x^2-14x+4}{(x^2+2)^4}$. [5]

$y = \frac{x^2+2x+3}{(x^2+2)^3}$

which is in the form $y = \frac{u}{v}$.

Let $u = x^2 + 2x + 3$ and $v = (x^2 + 2)^3$

$\frac{du}{dx} = 2x + 2$

$\frac{dv}{dx} = 3(2x)(x^2 + 2)^{3-1}$

$\frac{dv}{dx} = 6x(x^2 + 2)^2$

Using the quotient rule,

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

$$= \frac{(x^2+2)^3(2x+2) - (x^2+2x+3)(6x)(x^2+2)^2}{[(x^2+2)^3]^2}$$

$$= \frac{(x^2+2)^2[(x^2+2)(2x+2)-(x^2+2x+3)(6x)]}{(x^2+2)^2\cdot(x^2+2)^4}$$

$$= \frac{(x^2+2)(2x+2)-(x^2+2x+3)(6x)}{(x^2+2)^4}$$

$$= \frac{2x^3+2x^2+4x+4-6x^3-12x^2-18x}{(x^2+2)^4}$$

$$= \frac{-4x^3-10x^2-14x+4}{(x^2+2)^4}$$

$$\therefore \frac{dy}{dx} = \frac{-4x^3-10x^2-14x+4}{(x^2+2)^4}$$

Q.E.D.

(c) The equation of an ellipse is given by

$$x = 1 - 3\cos\theta, \qquad y = 2\sin\theta, \quad 0 \le \theta \le 2\pi.$$

Find $\frac{dy}{dx}$ in terms of $\theta$. [5]

$x = 1 - 3\cos\theta$ , $y = 2\sin\theta$

$\frac{dx}{d\theta} = -3(-\sin\theta)$ $\frac{dy}{d\theta} = 2\cos\theta$

$\frac{dx}{d\theta} = 3\sin\theta$

Using the chain rule,

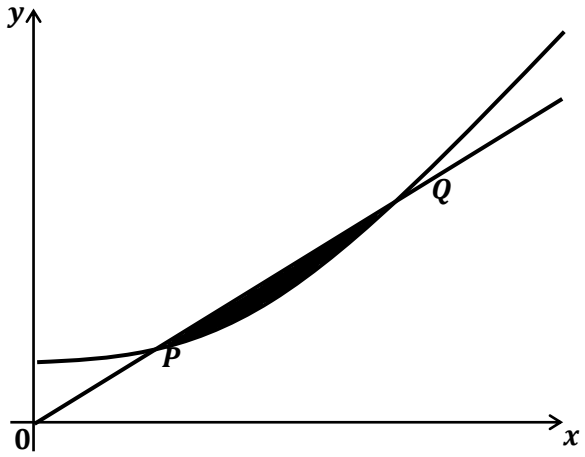
$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= 2\cos\theta \times \frac{1}{3\sin\theta}$$

$$= \frac{2}{3} \cdot \frac{\cos\theta}{\sin\theta}$$

$$= \frac{2}{3}\cot\theta$$

(d) The diagram below (**not drawn to scale**) shows the curve $y = x^2 + 3$ and the line $y = 4x$.




(i) Determine the coordinates of the points $\boldsymbol{P}$ and $\boldsymbol{Q}$ at which the curve and the line intersect. [4]

$y = x^2 + 3$ → Equation 1

$y = 4x$ → Equation 2

Substituting Equation 2 into Equation 1 gives:

$$4x = x^2 + 3$$

$$x^2 - 4x + 3 = 0$$

$$(x - 1)(x - 3) = 0$$

Either $x - 1 = 0$ or $x - 3 = 0$

$x = 1$ $x = 3$

When $x = 1$,

$y = 4(1)$

$= 4$

When $x = 3$,

$y = 4(3)$

$= 12$

$\therefore$ The coordinates are $P(1, 4)$ and $Q(3, 12)$.

(ii) Calculate the area of the shaded region. [5]

The equation of the line is:

$y = 4x$

So, we have,

Area under line $= \int_a^b y\,dx$

$= \int_1^3 4x\,dx$

$= \left[\frac{4x^2}{2}\right]_1^3$

$= [2x^2]_1^3$

$= 2(3)^2 - 2(1)^2$

$= 2(9) - 2$

$= 18 - 2$

$= 16 \text{ units}^2$

The equation of the curve is:

$y = x^2 + 3$

Area under curve $= \int_a^b y\,dx$

$= \int_1^3 (x^2 + 3)\,dx$

$= \left[\frac{x^3}{3} + 3x\right]_1^3$

$= \left[\frac{(3)^3}{3} + 3(3)\right] - \left[\frac{(1)^3}{3} + 3(1)\right]$

$= 9 + 9 - \left(\frac{1}{3} + 3\right)$

$= 18 - \frac{10}{3}$

$= \frac{44}{3}$ units$^2$

Therefore,

Area of the shaded region = Area under the line − Area under the curve

$= 16 - \frac{44}{3}$

$= \frac{4}{3}$ units$^2$

**Total: 25 marks**

June 2013 – Question 6 – Solution

(a) (i) By using the substitution $u = 1 - x$, find $\int x(1-x)^2 \ dx$. [5]

Let $u = 1 - x$

$\frac{du}{dx} = -1$

$dx = -du$

Also, $x = 1 - u$.

Now,

$$\int x(1-x)^2 \ dx = \int (1-u)u^2 \, (-1) \, du$$

$$= \int (-1)(u^2 - u^3) \ du$$

$$= \int (u^3 - u^2) \ du$$

$$= \frac{u^4}{4} - \frac{u^3}{3} + c$$

$$= \frac{(1-x)^4}{4} - \frac{(1-x)^3}{3} + c$$

$$\therefore \int x(1-x)^2 \ dx = \frac{(1-x)^4}{4} - \frac{(1-x)^3}{3} + c$$

(ii) Given that $f(t) = 2\cos t, \quad g(t) = 4\sin 5t + 3\cos t$

show that $\int [f(t) + g(t)]\, dt = \int f(t)\, dt + \int g(t)\, dt$. [4]

Taking L.H.S:

$$\int [f(t) + g(t)]\, dt = \int (2\cos t + 4\sin 5t + 3\cos t)\, dt$$

$$= \int (5\cos t + 4\sin 5t)\, dt$$

$$= 5\sin t - \frac{4}{5}\cos 5t + c$$

Taking R.H.S:

$$\int f(t)\, dt + \int g(t)\, dt = \int 2\cos t\, dt + \int (4\sin 5t + 3\cos t)\, dt$$

$$= 2\sin t - \frac{4}{5}\cos 5t + 3\sin t + c$$

$$= 5\sin t - \frac{4}{5}\cos 5t + c$$

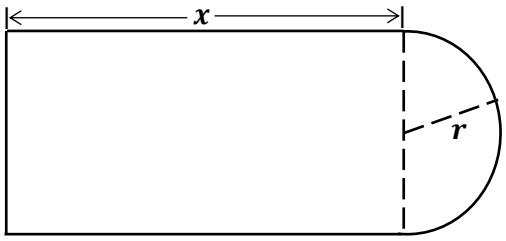
Since $\int [f(t) + g(t)]\, dt = 5\sin t - \frac{4}{5}\cos 5t + c$

and $\int f(t)\, dt + \int g(t)\, dt = 5\sin t - \frac{4}{5}\cos 5t + c$,

then $\int [f(t) + g(t)]\, dt = \int f(t)\, dt + \int g(t)\, dt$.

Q.E.D.

(b) A sports association is planning to construct a running track in the shape of a rectangle surmounted by a semicircle, as shown in the diagram below. The letter $x$ represents the length of the rectangular section and $r$ represents the radius of the semicircle.




The perimeter of the track must be 600 metres.

(i) Show that $r = \frac{600-2x}{2+\pi}$. [2]

Perimeter of track $= 2x + 2r + \frac{1}{2}(2\pi r)$

Since the perimeter of the track is 600 $m$, we have

$$600 = 2x + 2r + \frac{1}{2}(2\pi r)$$

$$600 = 2x + 2r + \pi r$$

$$2r + \pi r = 600 - 2x$$

$$r(2 + \pi) = 600 - 2x$$

$$r = \frac{600-2x}{2+\pi}$$

$$\therefore r = \frac{600-2x}{2+\pi}$$

(ii) Hence, determine the length, $x$, that **maximises** the area enclosed by the track. [6]

Let $A$ be the area of the track.

$A =$ Area of rectangle + Area of semicircle

$$= 2rx + \frac{1}{2}\pi r^2$$

$$= 2x\left(\frac{600-2x}{2+\pi}\right) + \frac{1}{2}\pi\left(\frac{600-2x}{2+\pi}\right)^2$$

$$= \frac{2}{2+\pi}(600x - 2x^2) + \frac{\pi}{2(2+\pi)^2}(600-2x)^2$$

Now,

$$\frac{dA}{dx} = \frac{2}{2+\pi}(600-4x) + \frac{\pi}{2(2+\pi)^2} \times 2(-2)(600-2x)$$

$$= \frac{1200-8x}{2+\pi} + \frac{2(-2\pi)(600-2x)}{2(\pi+2)^2}$$

$$= \frac{1200-8x}{2+\pi} - \frac{2\pi(600-2x)}{(\pi+2)^2}$$

When $\frac{dA}{dx} = 0$,

$$\frac{1200-8x}{2+\pi} - \frac{2\pi(600-2x)}{(\pi+2)^2} = 0$$

$\times (2+\pi)^2$

$$(1200-8x)(2+\pi) = 2\pi(600-2x)$$

$$2400 + 1200\pi - 16x - 8x\pi = 1200\pi - 4\pi x$$

$$2400 = 16x + 4\pi x$$

$$2400 = x(16+4\pi)$$

$$x = \frac{2400}{16+4\pi}$$

$$x = 84\ m$$

Now,

$$\frac{d^2A}{dx^2} = \frac{-8}{2+\pi} + \frac{4\pi}{(2+\pi)^2}$$

$$= \frac{-8(2+\pi)+4\pi}{(2+\pi)^2}$$

$$= \frac{-16-8\pi+4\pi}{(2+\pi)^2}$$

$$= \frac{-16-4\pi}{(2+\pi)^2}$$ which is less than 0.

$\therefore x = 84\ m$ gives a maximum area.

(c) (i) Let $y = -x\sin x - 2\cos x + Ax + B$, where $A$ and $B$ are constants.

Show that $y'' = x\sin x$. [4]

$$y = -x\sin x - 2\cos x + Ax + B$$

$$y' = (-x)(\cos x) + (-1)(\sin x) - 2(-\sin x) + A$$

$$= -x\cos x - \sin x + 2\sin x + A$$

$$= -x\cos x + \sin x + A$$

Now,

$$y'' = (-x)(-\sin x) + (-1)(\cos x) + \cos x$$

$$= x\sin x - \cos x + \cos x$$

$$= x\sin x$$

$\therefore y'' = x\sin x$

Q.E.D.

(ii) Hence, determine the specific solution of the differential equation

$$y'' = x \sin x,$$

given that when $x = 0$, $y = 1$ and when $x = \pi$, $y = 6$. [4]

$y" = x \sin x$

$y = -x \sin x - 2 \cos x + Ax + B$

When $x = 0, y = 1,$

$1 = -(0) \sin(0) - 2 \cos(0) + A(0) + B$

$1 = 0 - 2(1) + 0 + B$

$1 = -2 + B$

$B = 1 + 2$

$B = 3$

When $x = \pi, y = 6$ and $B = 3,$

$6 = -(\pi) \sin(\pi) - 2 \cos(\pi) + A(\pi) + 3$

$6 = -(\pi)(0) - 2(-1) + A(\pi) + 3$

$6 = 2 + \pi A + 3$

$6 = 5 + \pi A$

$\pi A = 6 - 5$

$\pi A = 1$

$A = \frac{1}{\pi}$

$\therefore y = -x \sin x - 2 \cos x + \frac{x}{\pi} + 3$ (P.S.)

**Total: 25 marks**